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一、配置

开启RocksDB需要注意前置的一些配置。

参数	值	含义
state.backend	rocksdb	
state.backend.async	true	Flink 1.13后默认开启异步
state.backend.incremental	true	只针对 RocksDB 生效
state.backend.local-recovery	true	非TM问题，可以无需从 hdfs 拉取数据

EmbeddedRocksDBStateBackend 使用介绍。

需要注意的是，如果使用RocksDB作为状态后端，红框中的CheckpointStorage不能使用 JobManagerCheckpointStorage。

```
env.enableCheckpointing(checkPointInterval, CheckpointingMode.EXACTLY_ONCE);
env.getCheckpointConfig().setCheckpointTimeout(checkPointTimeout);
env.getCheckpointConfig().setCheckpointStorage(new JobManagerCheckpointStorage());
env.setStateBackend(new EmbeddedRocksDBStateBackend());
env.getConfig().enableObjectReuse();
```

否则会报如下错误，cp检查点数据大小会大于默认5MB，job失败。

```
11791 at java.util.concurrent.InnreadPoolExecutor$Worker.run(InnreadPoolExecutor.java:64) [?:1.8.0_332]
71792 at java.lang.Thread.run(Thread.java:750) [?:1.8.0_332]
71793 Caused by: org.apache.flink.util.SerializedThrowable: Could not materialize checkpoint 1 for operator Flat Map (181/300)#0.
71794 at org.apache.flink.streaming.runtime.tasks.AsyncCheckpointRunnable.handleExecutionException(AsyncCheckpointRunnable.java:257) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71795 ... 4 more
71796 Caused by: org.apache.flink.util.SerializedThrowable: java.io.IOException: Size of the state is larger than the maximum permitted memory-backed state. Size=65039762, maxSize=5242880. Consider using a
storage, like the FileSystemCheckpointStorage.
71797 at java.util.concurrent.FutureTask.report(FutureTask.java:122) ~[?:1.8.0_332]
71798 at java.util.concurrent.FutureTask.get(FutureTask.java:192) ~[?:1.8.0_332]
71799 at org.apache.flink.runtime.concurrent.FutureUtils.runIfNotDoneAndGet(FutureUtils.java:636) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71800 at org.apache.flink.streaming.api.operators.OperatorSnapshotFinalizer.<init>(OperatorSnapshotFinalizer.java:54) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71801 at org.apache.flink.streaming.runtime.tasks.AsyncCheckpointRunnable.run(AsyncCheckpointRunnable.java:128) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71802 ... 3 more
71803 Caused by: org.apache.flink.util.SerializedThrowable: Size of the state is larger than the maximum permitted memory-backed state. Size=65039762, maxSize=5242880. Consider using a different checkpoint
FileSystemCheckpointStorage.
71804 at org.apache.flink.runtime.state.memory.MemCheckpointStreamFactory.checkSize(MemCheckpointStreamFactory.java:63) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71805 at org.apache.flink.runtime.state.memory.MemCheckpointStreamFactory$MemoryCheckpointOutputStream.closeAndGetBytes(MemCheckpointStreamFactory.java:140) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71806 at org.apache.flink.runtime.state.memory.MemCheckpointStreamFactory$MemoryCheckpointOutputStream.closeAndGetHandle(MemCheckpointStreamFactory.java:120) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71807 at org.apache.flink.contrib.streaming.state.RocksDBStateUploader.uploadLocalFileToCheckpointFs(RocksDBStateUploader.java:143) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71808 at org.apache.flink.contrib.streaming.state.RocksDBStateUploader.lambda$createUploadFutures$0(RocksDBStateUploader.java:101) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71809 at org.apache.flink.util.Function.CheckedSupplier.lambda$unchecked$0(CheckedSupplier.java:32) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71810 at java.util.concurrent.CompletableFuture$AsyncSupply.run(CompletableFuture.java:1604) ~[?:1.8.0_332]
71811 ... 3 more
71812 2024-02-23 14:52:03,024 [flink-akka.actor.default-dispatcher-165] INFO org.apache.flink.runtime.jobmaster.JobMaster [] - Trying to recover from a global failure.
71813 org.apache.flink.util.FlinkRuntimeException: Exceeded checkpoint tolerable failure threshold.
71814 at org.apache.flink.runtime.checkpoint.CheckpointFailureManager.handleTaskLevelCheckpointException(CheckpointFailureManager.java:98) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71815 at org.apache.flink.runtime.checkpoint.CheckpointFailureManager.handleTaskLevelCheckpointException(CheckpointFailureManager.java:84) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71816 at org.apache.flink.runtime.checkpoint.CheckpointCoordinator.abortPendingCheckpoint(CheckpointCoordinator.java:1931) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
71817 at org.apache.flink.runtime.checkpoint.CheckpointCoordinator.receiveDeclineMessage(CheckpointCoordinator.java:991) ~[flink-dist_2.12-1.13.6.jar:1.13.6]
```

实际上checkpoint的状态主要有两种 (<https://www.cnpython.com/java/383457>)。

1. 存储工作状态的位置（状态后端）
2. 存储检查点的位置（检查点存储）

在使用了rocksDB后，检查点状态存储量级增大，此时JM作为存储点回超过限制报错。

```
// flink-conf.yaml中配置
state.checkpoint-storage: filesystem
//代码中指定
env.getCheckpointConfig().setCheckpointStorage(new FileSystemCheckpointStorage("file:///checkpoint-dir"));
```

其他配置

```
2 MapStateDescriptor<String, byte[]> mapStateDescriptor =
3     new MapStateDescriptor<>("mapstate",
4         TypeInformation.of(String.class),
5         TypeInformation.of(byte[].class));
6 MapStateDescriptor<String, byte[]> bidMapStateDescriptor =
7     new MapStateDescriptor<>("bidMapstate",
8         TypeInformation.of(String.class),
9         TypeInformation.of(byte[].class));
10
11 // 过期时间10分钟,
12 StateTtlConfig ttlConfigString = StateTtlConfig
13     .newBuilder(Time.seconds(600))
14     .setUpdateType(StateTtlConfig.UpdateType.OnCreateAndWrite)
15     .setStateVisibility(StateTtlConfig.StateVisibility.NeverReturnExpired).build();
16
17 // 存储曝光
18 mapStateDescriptor.enableTimeToLive(ttlConfigString);
19 mapState = getRuntimeContext().getMapState(mapStateDescriptor);
20
21 // 存储竞价
22 bidMapStateDescriptor.enableTimeToLive(ttlConfigString);
23 bidMapState = getRuntimeContext().getMapState(bidMapStateDescriptor);
```

原理性介绍

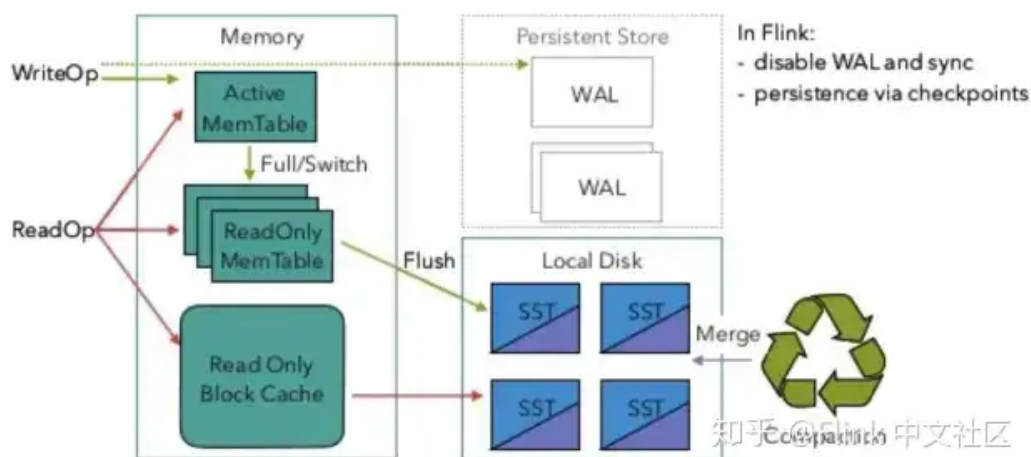
<https://zhuanlan.zhihu.com/p/256351671>

<https://www.jianshu.com/p/569a7e67c1b3>

<https://zhuanlan.zhihu.com/p/616209332>

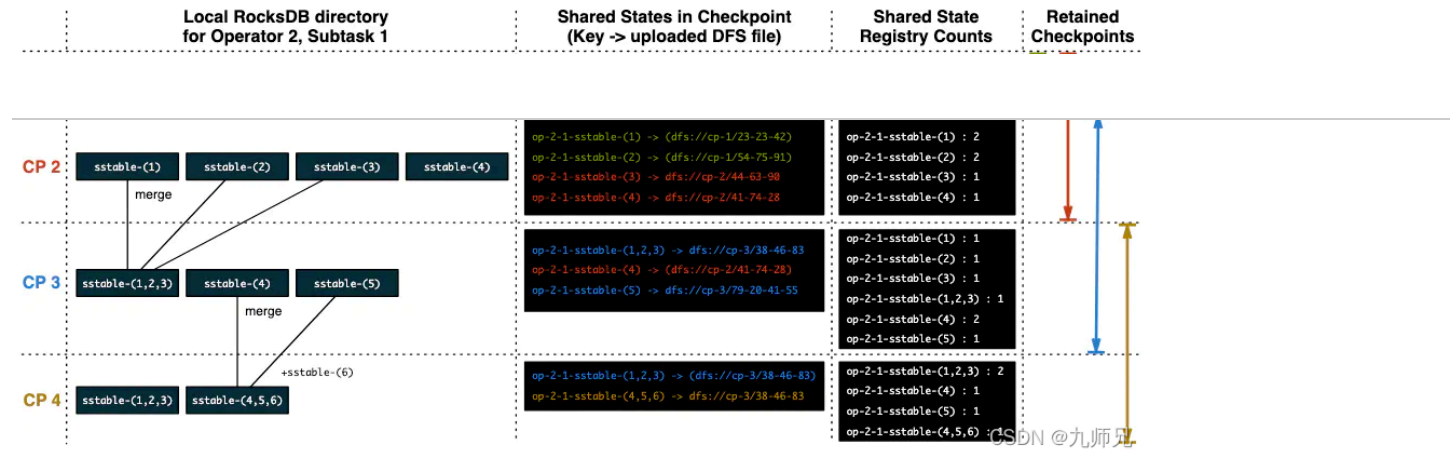
https://blog.csdn.net/qq_21383435/article/details/122258048

TM + LocalDisk视角



checkpoint视角

主要说明了RocksDB的sstable 如何 关联 hdfs的目录文件



二、测试效果

首先10%采样数据

cp时长平均20s, rocksDB后端状态大概稳定在23GB左右

rtlj

CANCELED

4

ID: b59eefed580131e9e62b5e1137f2493f

Start Time: 2024-02-26 15:10:47

End Time: 2024-02-26 16:08:24

Duration: 57m 37s

Overview

Exceptions

TimeLine

Checkpoints

Configuration

Overview

History

Summary

Configuration

	ID	Status	Acknowledged	Trigger Time	Latest Acknowledgement	End to End Duration	Checkpointed Data Size	Processed (persisted) in-flight d
+	11	COMPLETED	901/901	2024-02-26 16:03:24	2024-02-26 16:03:40	15s	23.3 GB	22.5 MB (0 B)
+	10	COMPLETED	901/901	2024-02-26 15:58:24	2024-02-26 15:58:39	14s	21.9 GB	31.4 MB (0 B)
+	9	COMPLETED	901/901	2024-02-26 15:53:24	2024-02-26 15:53:39	15s	23.5 GB	46.9 MB (0 B)
+	8	COMPLETED	901/901	2024-02-26 15:48:24	2024-02-26 15:48:41	17s	24.5 GB	90.0 MB (0 B)
+	7	COMPLETED	901/901	2024-02-26 15:43:24	2024-02-26 15:43:41	17s	26.8 GB	90.3 MB (0 B)
+	6	COMPLETED	901/901	2024-02-26 15:38:24	2024-02-26 15:38:51	26s	24.8 GB	151 MB (0 B)
+	5	COMPLETED	901/901	2024-02-26 15:33:24	2024-02-26 15:33:45	20s	23.0 GB	249 MB (0 B)
+	4	COMPLETED	901/901	2024-02-26 15:28:24	2024-02-26 15:28:42	18s	21.7 GB	298 MB (0 B)
+	3	COMPLETED	901/901	2024-02-26 15:23:24	2024-02-26 15:23:44	19s	21.3 GB	222 MB (0 B)
+	2	COMPLETED	901/901	2024-02-26 15:18:24	2024-02-26 15:18:37	12s	12.8 GB	291 MB (0 B)

最终hdfs目录的大小稳定在80G左右

```
MVAD: [zhuming3@qt3ss] ~$ hdfs dfs -du -h viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/b59eefed580131e9e62b5e1137f2493f
4.5 M    13.4 M viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/b59eefed580131e9e62b5e1137f2493f/chk-4
27.8 G   83.3 G viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/b59eefed580131e9e62b5e1137f2493f/shared
0         0      viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/b59eefed580131e9e62b5e1137f2493f/taskowned
MVAD: [zhuming3@qt3ss] ~$ hive
```

无采样, 100%数据

cp时长平均80s, rocksDB后端状态大概稳定在90GB左右

Checkpointing History								
Overview History Summary Configuration								
	ID	Status	Acknowledged	Trigger Time	Latest Acknowledgement	End to End Duration	Checkpointed Data Size	P
+	13	COMPLETED	901/901	2024-02-26 17:11:27	2024-02-26 17:12:37	1m 10s	96.7 GB	2
+	12	COMPLETED	901/901	2024-02-26 17:06:27	2024-02-26 17:07:24	56s	56.2 GB	2
+	11	COMPLETED	901/901	2024-02-26 17:01:27	2024-02-26 17:02:43	1m 15s	72.5 GB	4
+	10	COMPLETED	901/901	2024-02-26 16:56:27	2024-02-26 16:57:35	1m 8s	86.7 GB	2
+	9	COMPLETED	901/901	2024-02-26 16:51:27	2024-02-26 16:52:50	1m 23s	87.3 GB	4
+	8	COMPLETED	901/901	2024-02-26 16:46:27	2024-02-26 16:47:32	1m 5s	57.4 GB	3
+	7	COMPLETED	901/901	2024-02-26 16:41:27	2024-02-26 16:42:34	1m 6s	76.6 GB	4
+	6	COMPLETED	901/901	2024-02-26 16:36:27	2024-02-26 16:38:07	1m 39s	104 GB	3
+	5	COMPLETED	901/901	2024-02-26 16:31:27	2024-02-26 16:32:42	1m 14s	97.0 GB	5
+	4	COMPLETED	901/901	2024-02-26 16:26:27	2024-02-26 16:28:20	1m 53s	154 GB	3

hdfs目录中大小, 稳定在500GB左右

```

MVAD: [zhuming3@qt3ss] ~$ hdfs dfs -du -h viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af450733
6.4 M      19.1 M      viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/chk
173.5 G    520.5 G    viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/sha
0          0          viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/tas
MVAD: [zhuming3@qt3ss] ~$ hdfs dfs -du -h viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af450733
8.6 M      25.8 M      viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/chk
149.9 G    449.7 G    viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/sha
0          0          viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/tas
MVAD: [zhuming3@qt3ss] ~$ hdfs dfs -du -h viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af450733
8.6 M      25.8 M      viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/chk
135.4 G    406.2 G    viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/sha
0          0          viewfs://ss-hadoop/apps/flink/shbt-k8s/checkpoints/bff2af45073364f53ee000428f995d22/tas

```

三、其他问题

在开启100%全量后，在MapState get操作，CPU占用明显偏高

Detail	SubTasks	TaskManagers	Watermarks	Accumulators	BackPressure	Metrics	FlameGraph
org.rocksdb.RocksDB.get:-2				org.rocksdb.RocksDB.get:-2			com.esoteric...
org.rocksdb.RocksDB.get:810				org.rocksdb.RocksDB.get:810			com.esotericof...
org.apache.flink.contrib.streaming.state.RocksD...				org.apache.flink.contrib.streaming.state...			com.esotericof...
org.apache.flink.runtime.state.ttl.TtlMapState.la...				org.apache.flink.runtime.state.ttl.TtlMap...			com.esotericof...
org.apache.flink.runtime.state.ttl.TtlMapState\$...				org.apache.flink.runtime.state.ttl.TtlMap...			com.esotericof...
org.apache.flink.runtime.state.ttl.AbstractTtlDec...				org.apache.flink.runtime.state.ttl.Abstra...			com.esotericsoft...
org.apache.flink.runtime.state.ttl.TtlMapState.ge...				org.apache.flink.runtime.state.ttl.TtlMap...			org.apache.flink...
org.apache.flink.runtime.state.ttl.TtlMapState.ge...				org.apache.flink.runtime.state.ttl.TtlMap...			org.apache.flink...
org.apache.flink.runtime.state.UserFacingMapSta...				org.apache.flink.runtime.state.UserFaci...		com.media...	org.apache.flink...
com.media.rtlj.topology.JoinerBolt.flatMap:127				com.media.rtlj.topology.JoinerBolt.flat...		com.media.rtlj...	org.apache.flink...
com.media.rtlj.topology.JoinerBolt.flatMap:33							org.apache.flink...
org.apache.flink.streaming.api.operators.StreamFlatMap.processElement:47							org.apache.flink...
org.apache.flink.streaming.runtime.tasks.OneInputStreamTask\$StreamTaskNetworkOutput.emitRecord:205							org.apache.flink...
org.apache.flink.streaming.runtime.io.AbstractStreamTaskNetworkInput.processElement:134							org.apache.flink.runtime.io.n...
org.apache.flink.streaming.runtime.io.AbstractStreamTaskNetworkInput.emitNext:105							org.apache.flink.streaming.runti...
org.apache.flink.streaming.runtime.io.StreamOneInputProcessor.processInput:66							
org.apache.flink.streaming.runtime.tasks.StreamTask.processInput:423							
org.apache.flink.streaming.runtime.tasks.StreamTask\$\$Lambda\$680/684516918.runDefaultAction:-1							
org.apache.flink.streaming.runtime.tasks.mailbox.MailboxProcessor.runMailboxLoop:204							
org.apache.flink.streaming.runtime.tasks.StreamTask.runMailboxLoop:684							
org.apache.flink.streaming.runtime.tasks.StreamTask.executeInvoke:639							
org.apache.flink.streaming.runtime.tasks.StreamTask\$\$Lambda\$1156/2067665827.run:-1							
org.apache.flink.streaming.runtime.tasks.StreamTask.runWithCleanUpOnFail:650							
org.apache.flink.streaming.runtime.tasks.StreamTask.invoke:623							
org.apache.flink.runtime.taskmanager.Task.doRun:779							

无标签